COMP108
Data Structures and Algorithms

Divide-and-Conquer Algorithms (Part I)

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Outline

Divide-and-Conquer algorithms

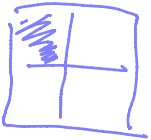
- Basic idea
- Learn a few examples

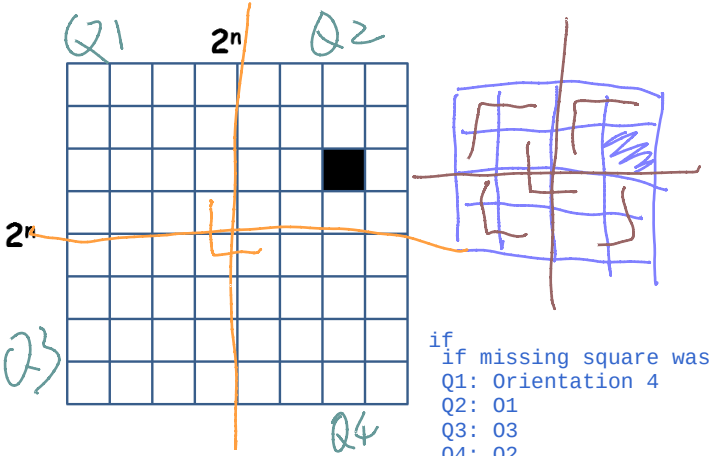
Learning outcomes:

- Able to describe the principle of divide-and-conquer algorithms
- Able to design divide-and-conquer algorithm for some simple problems

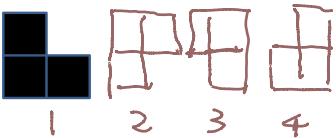
Triomino Puzzle

- **Input:** 2^n**-by-**2^n chessboard with **one** missing square & many **L-shaped** tiles of **3** adjacent squares
- **Question:** **Cover** the chessboard with L-shaped tiles without overlapping





Is it do-able?




```
if
 if missing square was in
 Q1: Orientation 4
 Q2: O1
 Q3: O3
 Q4: O2
```

Divide-and-conquer algorithms

One of the best-known algorithm design techniques

Idea:

- A problem instance is divided into several smaller instances of the same problem, ideally of about same size
- The smaller instances are solved, typically recursively
- The solutions for the smaller instances are combined to get a solution to the large instance

Some problems we have seen before

Finding the sum of all numbers in an array

- Suppose we have 8 numbers:

  `4 6 3 2 8 7 5 1`

Iterative version:

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sum ← 0
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while i ≤ n do
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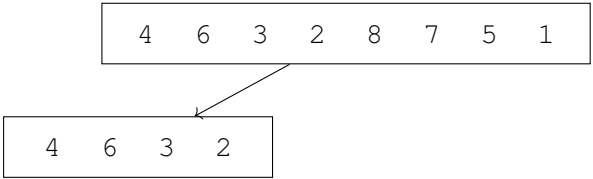

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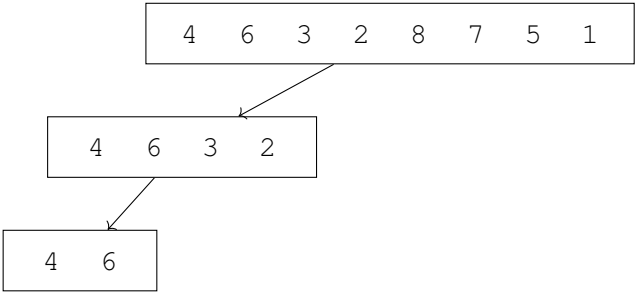

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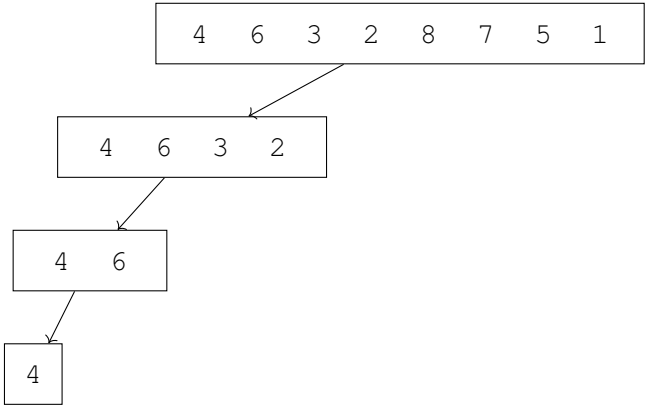

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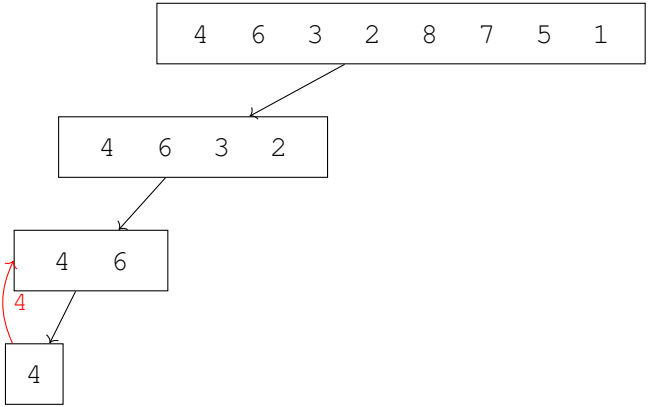

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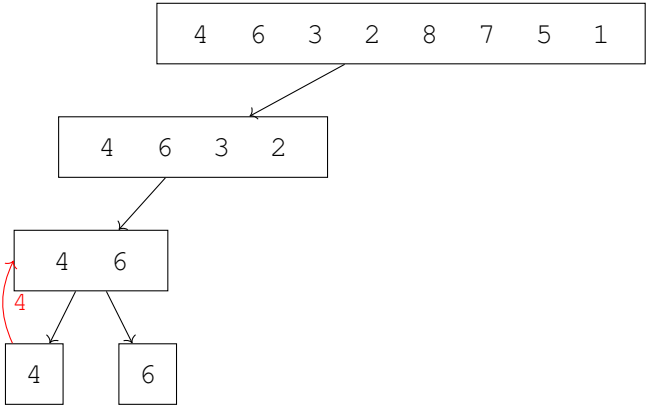

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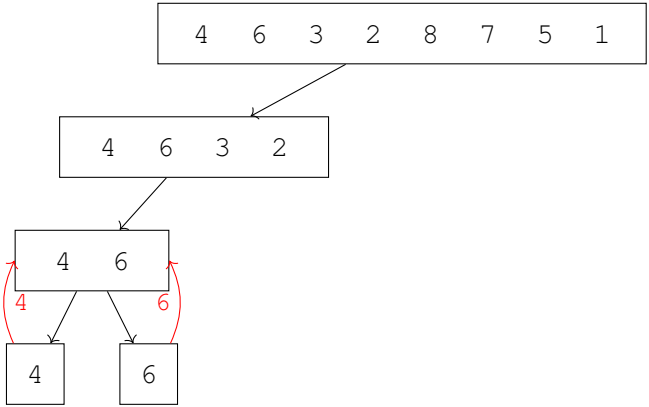

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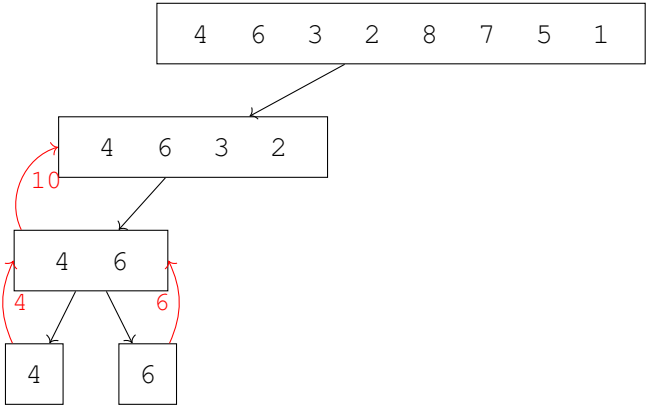

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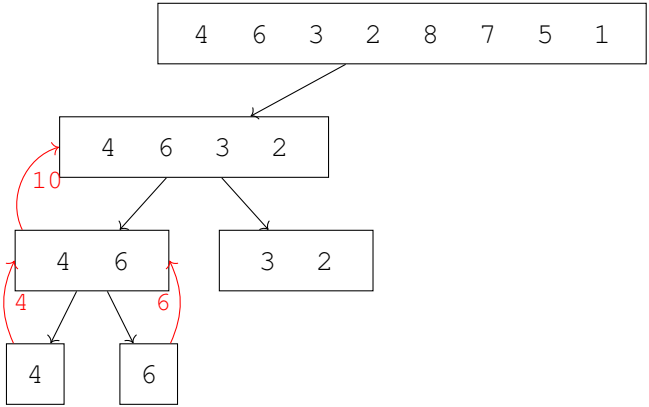

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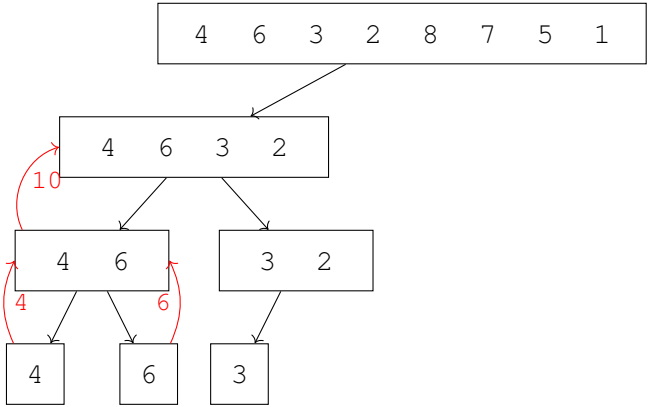

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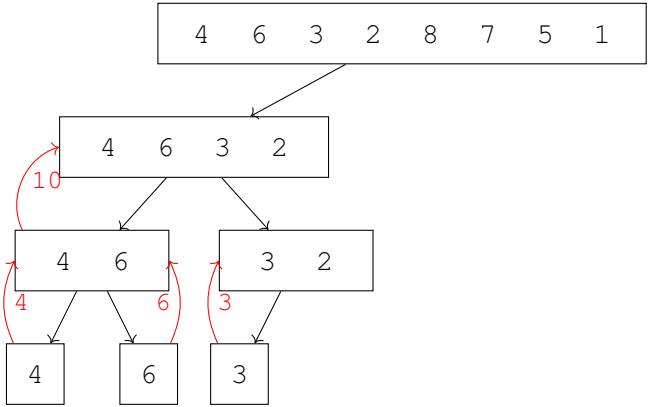

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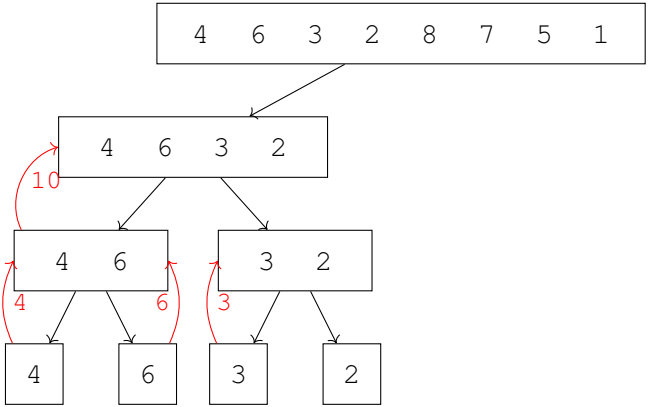

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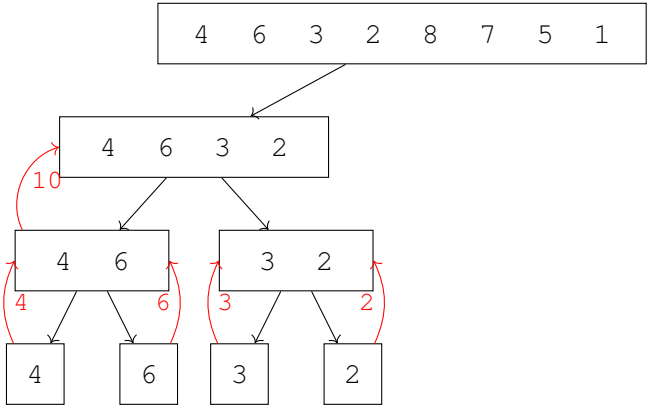

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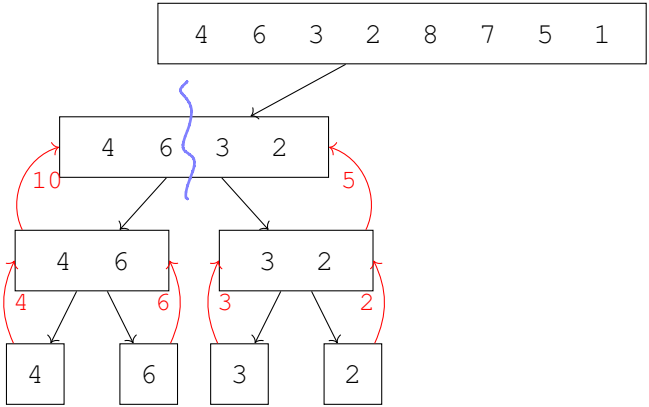

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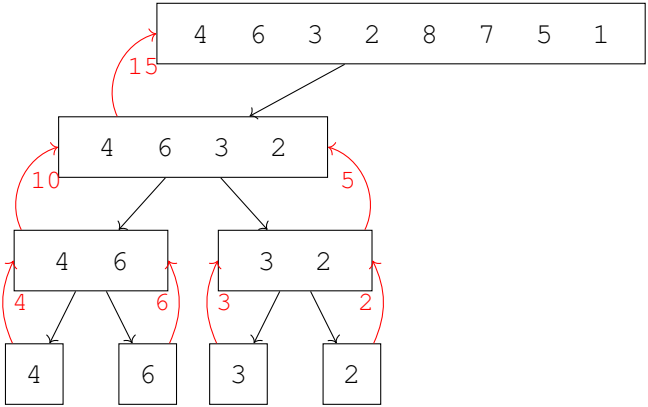

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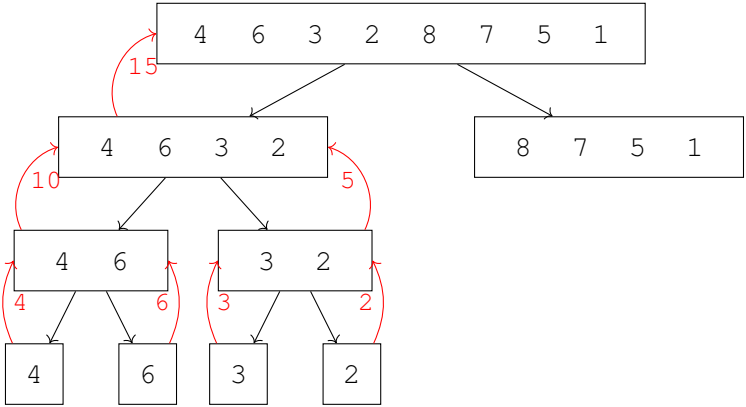

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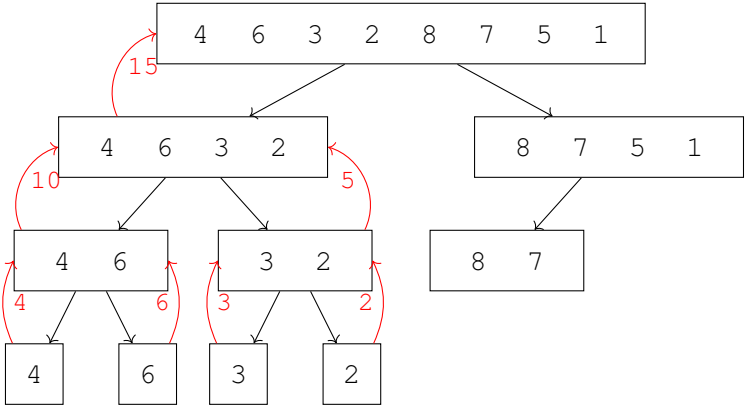

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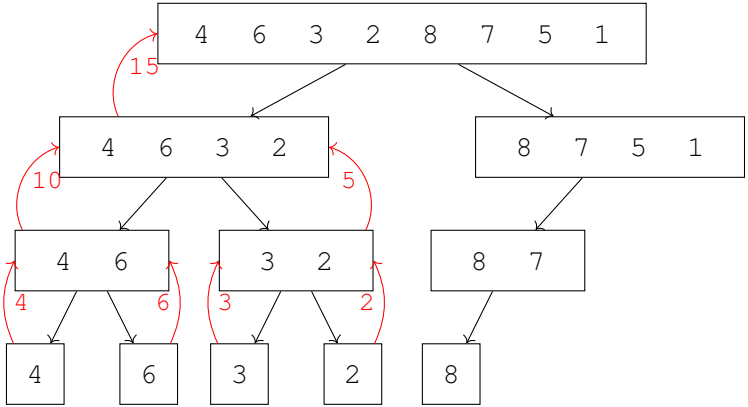

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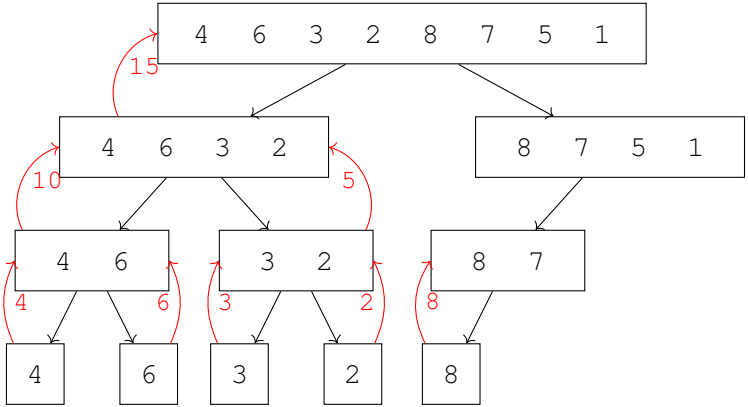

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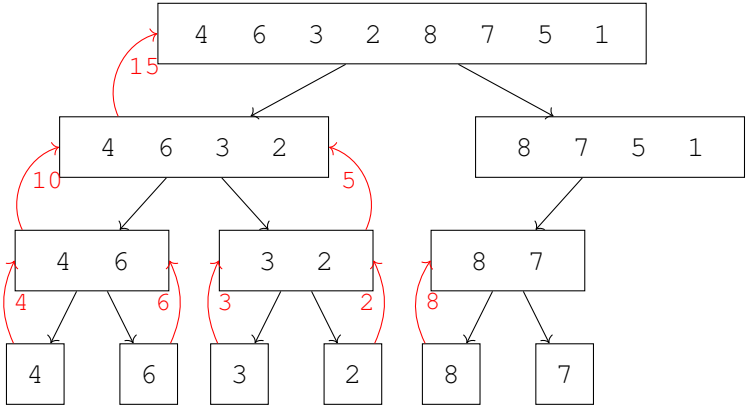

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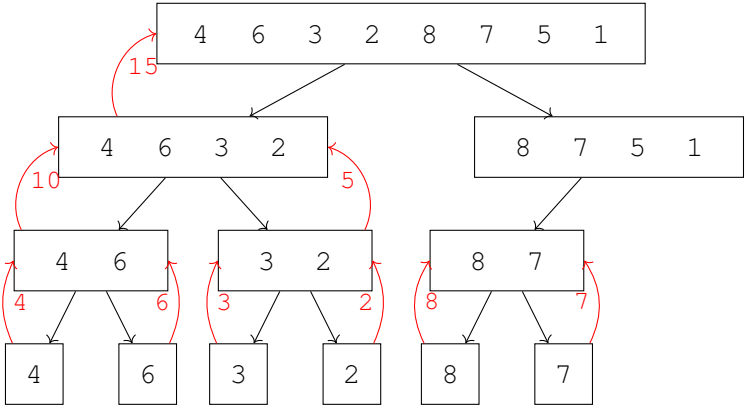

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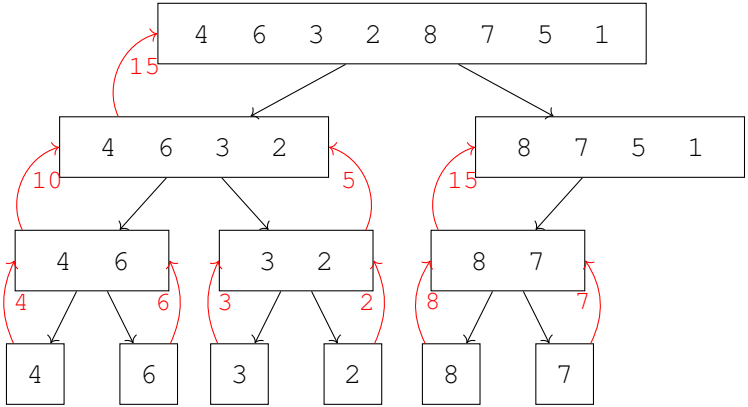

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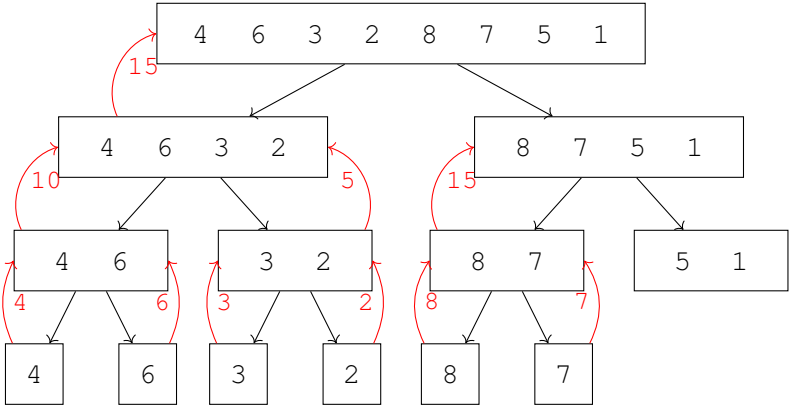

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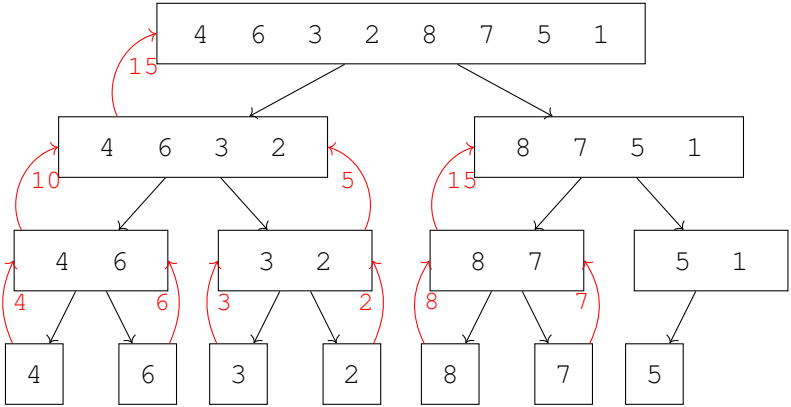

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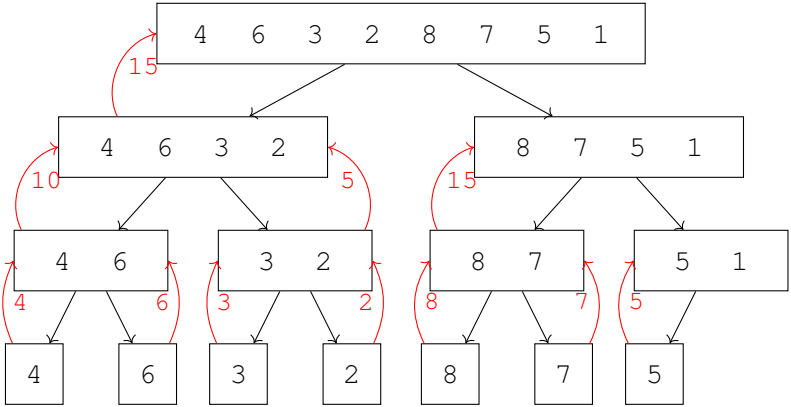

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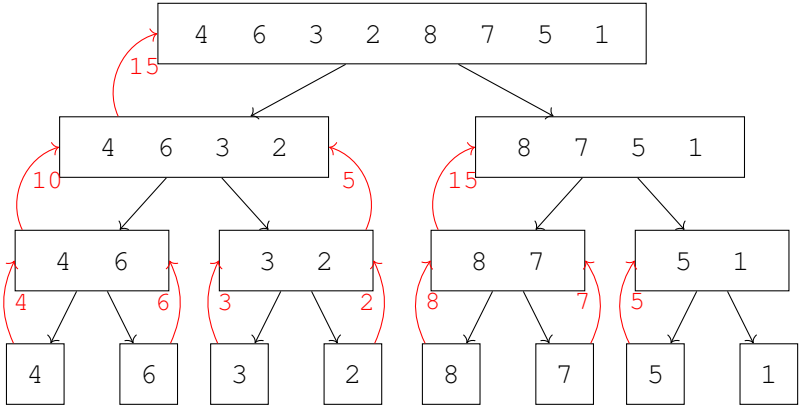

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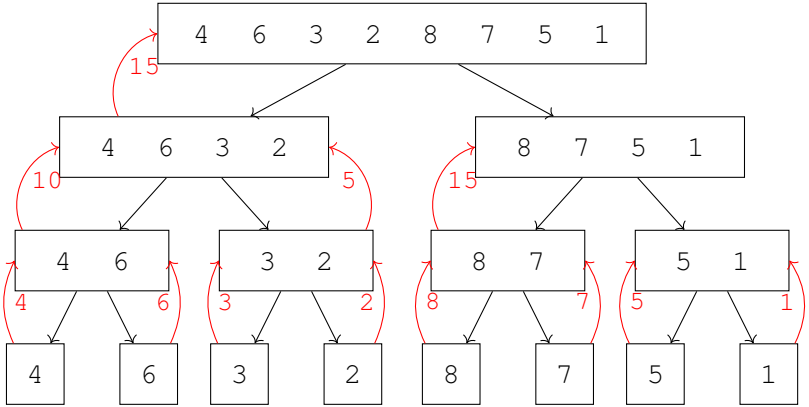

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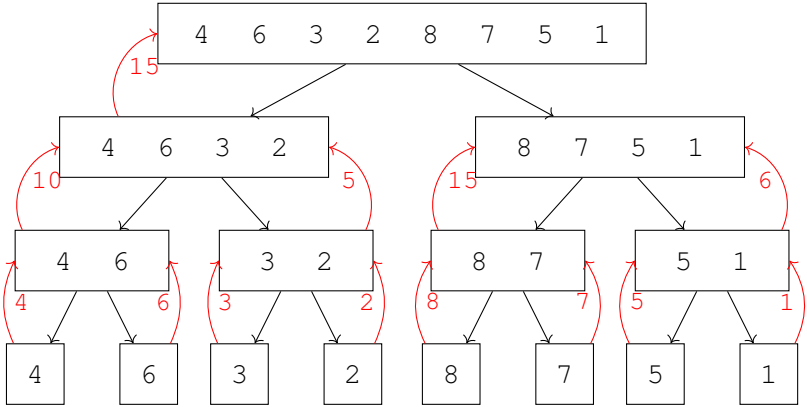

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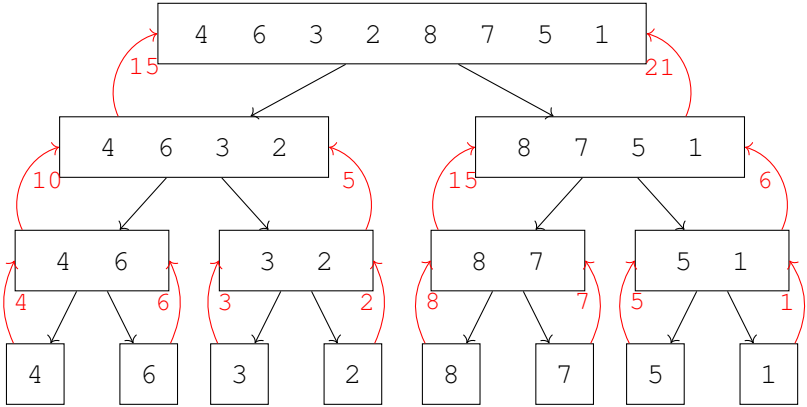

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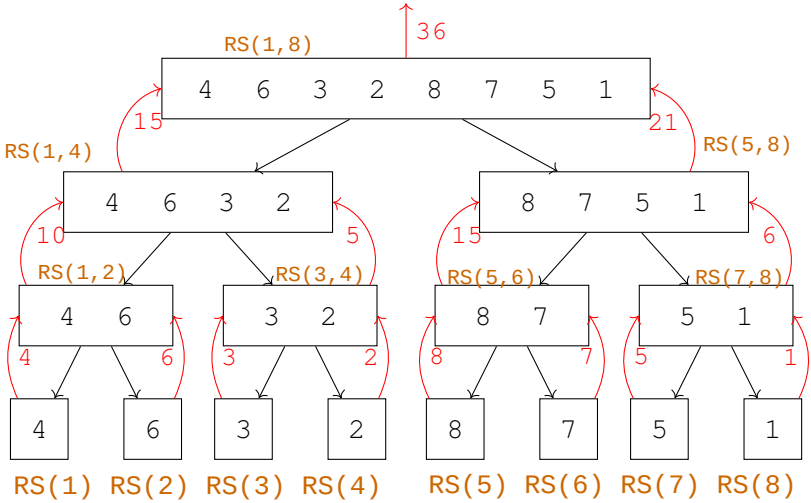

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We can call the following algorithm by RecSum(A, 1, n)

Algorithm RecSum($A[\,]$, p, q)

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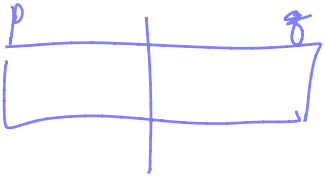

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        return sum1 + sum2
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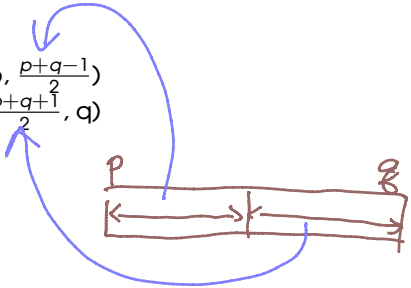

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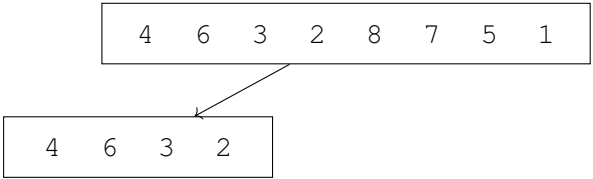

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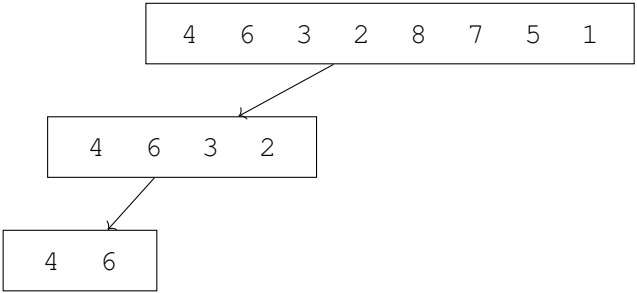

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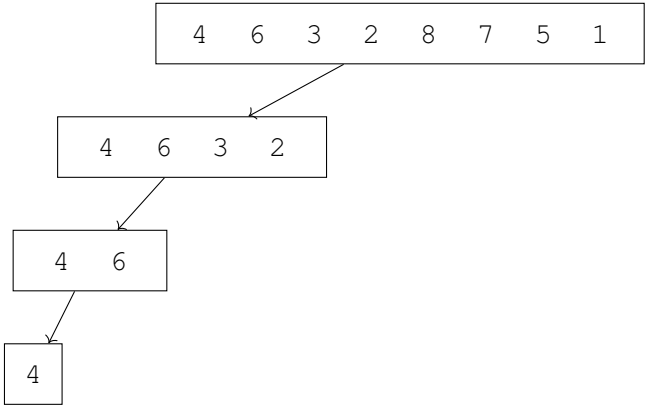

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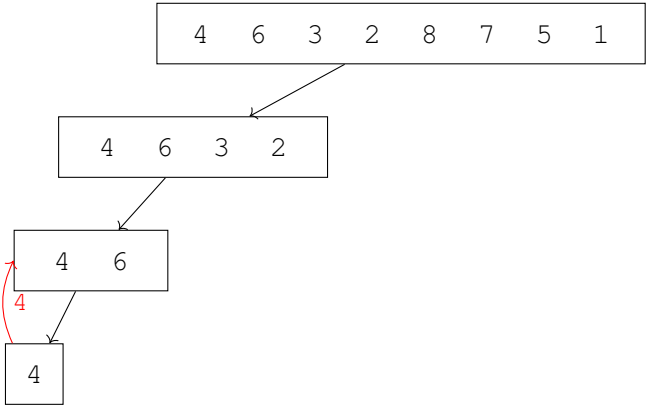

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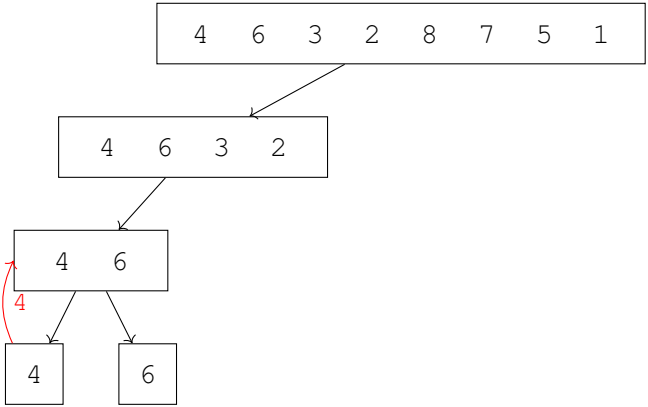

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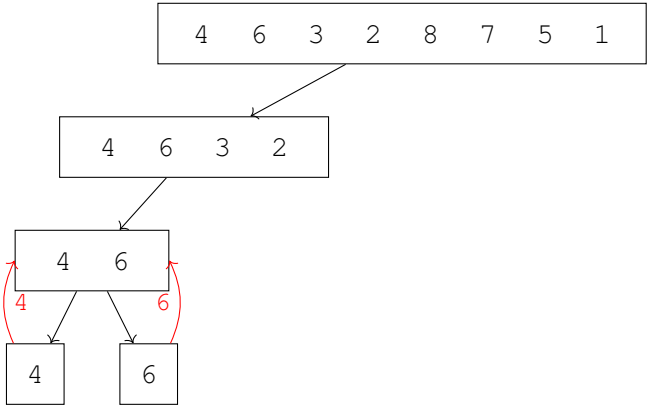

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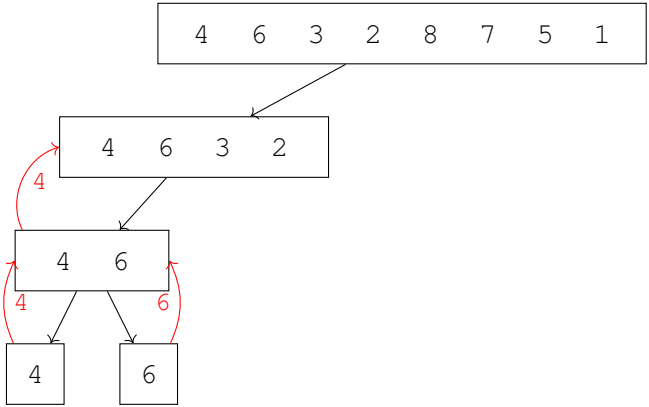

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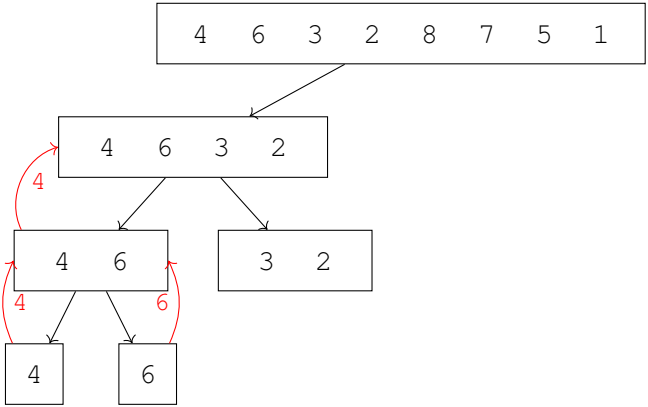

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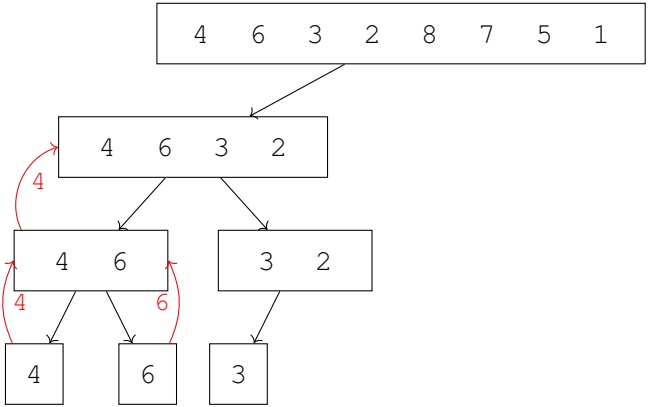

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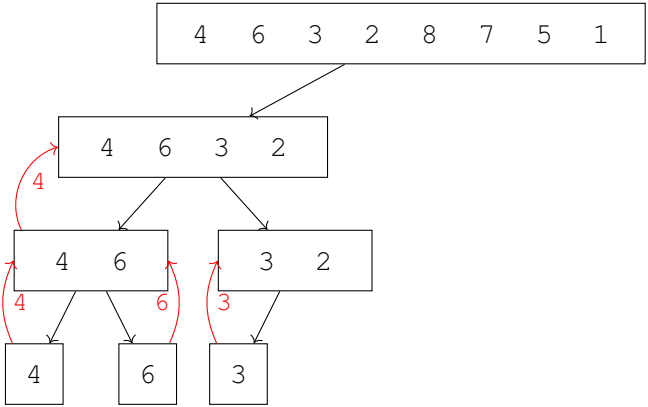

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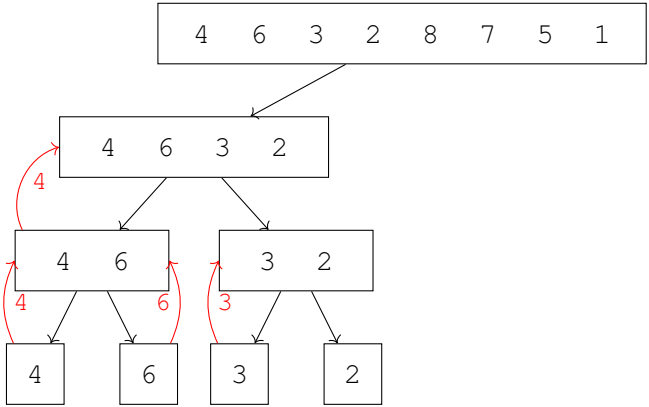

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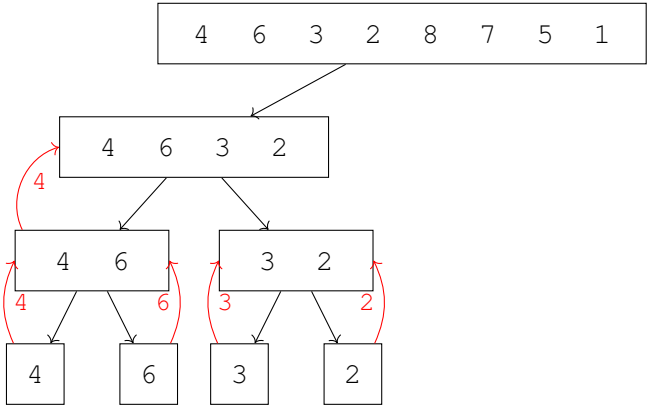

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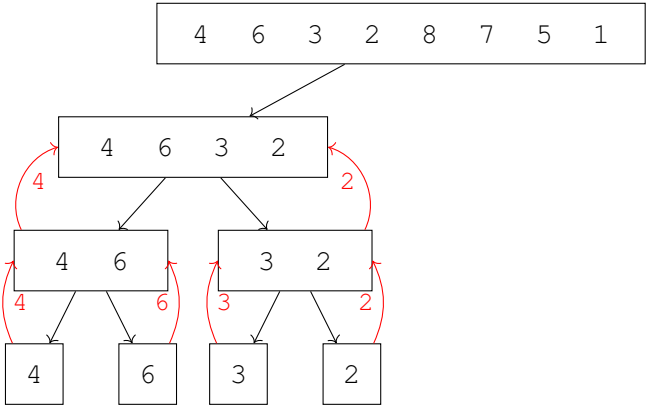

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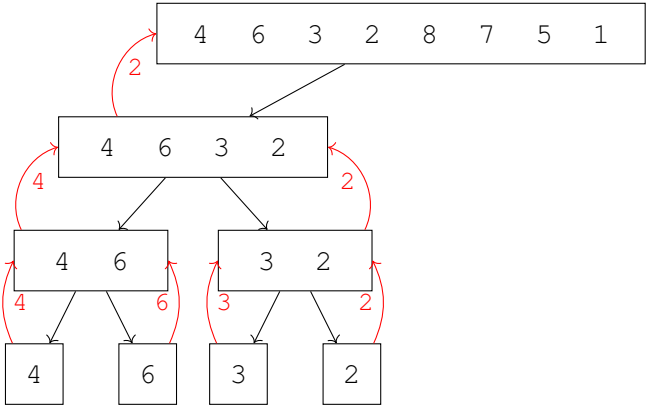

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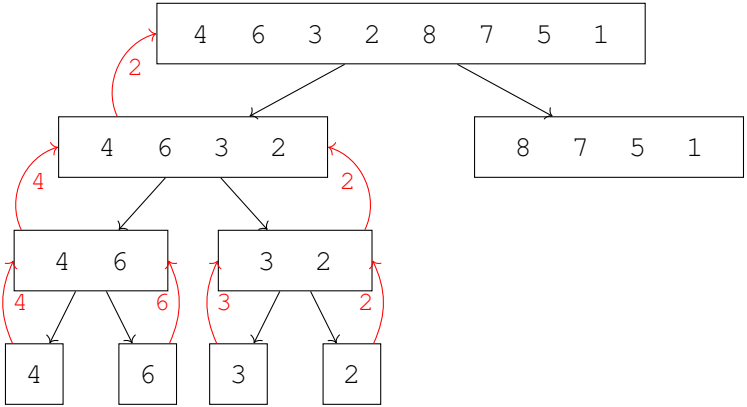

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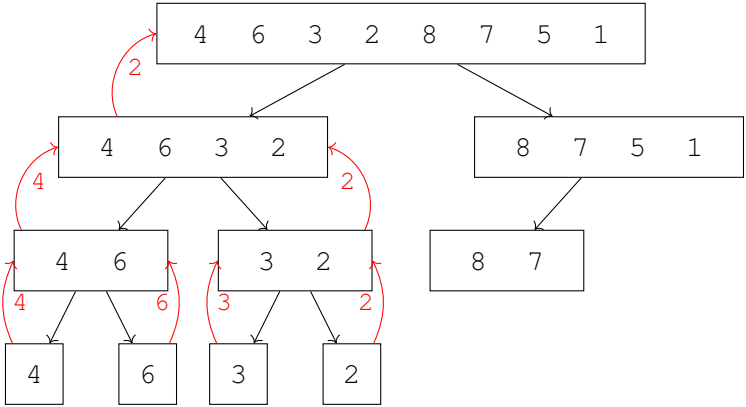

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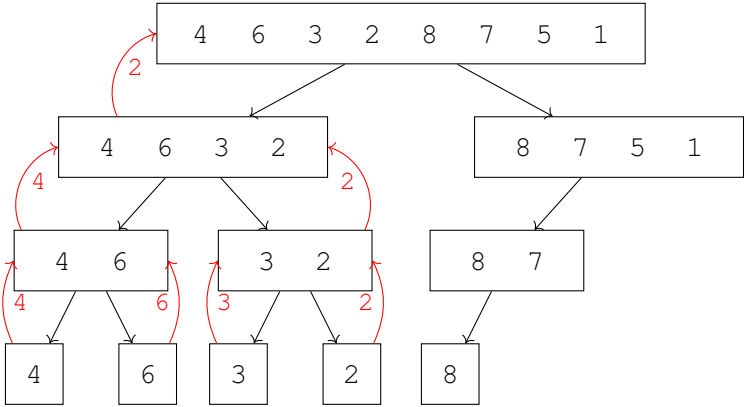

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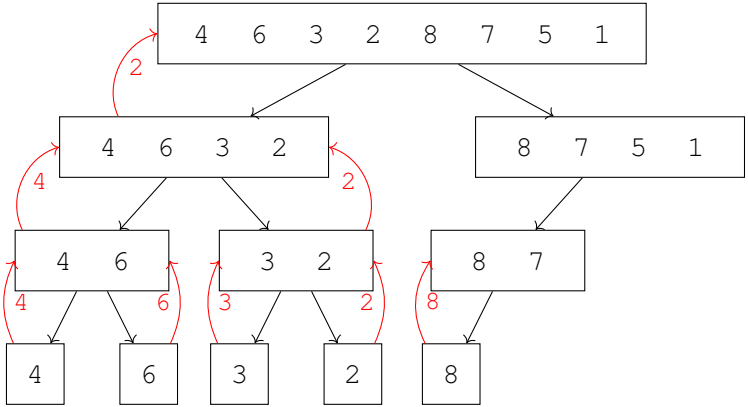

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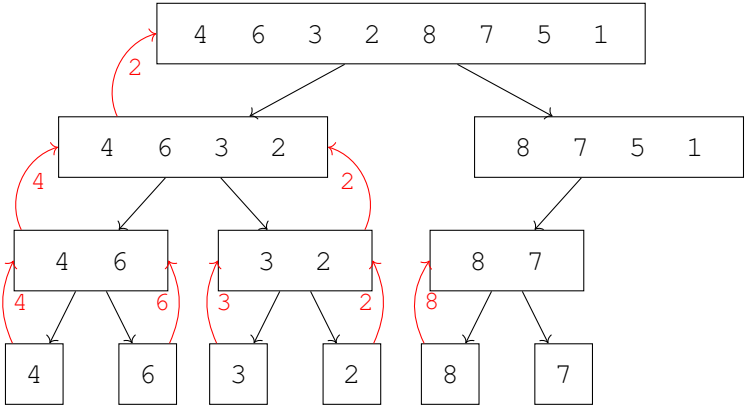

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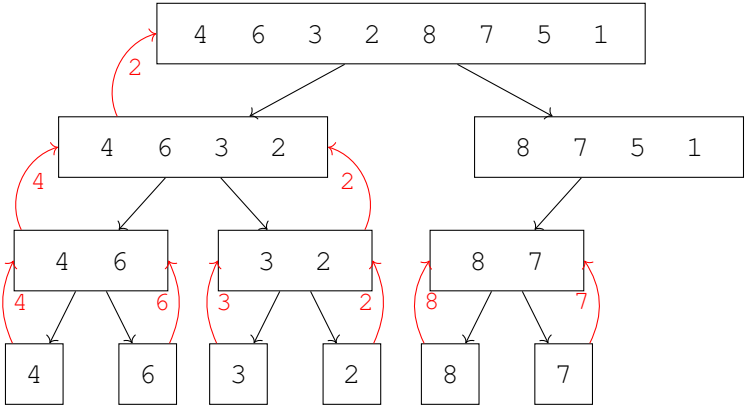

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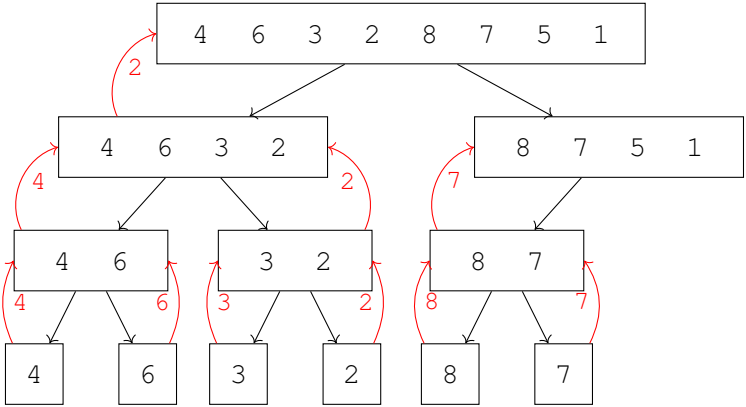

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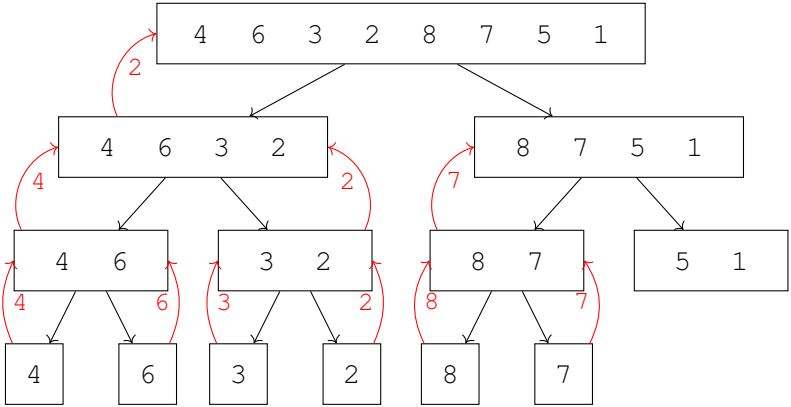

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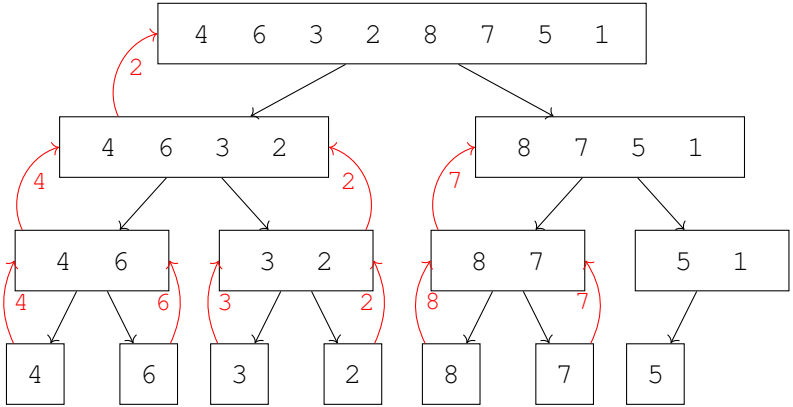

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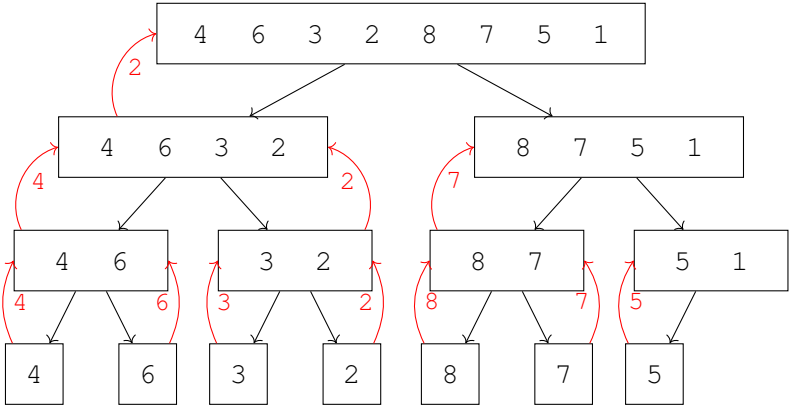

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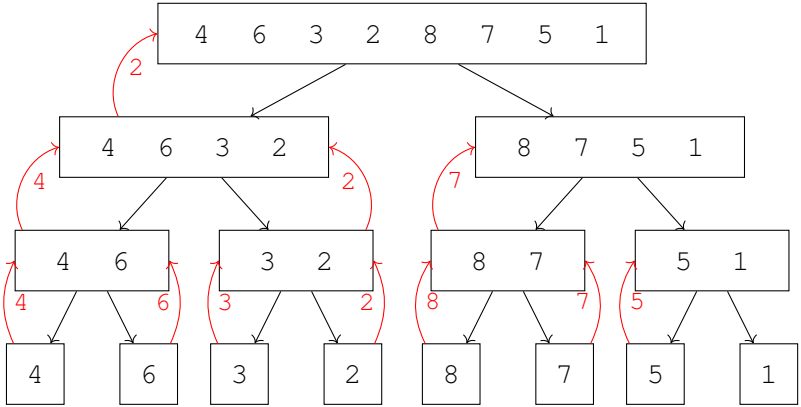

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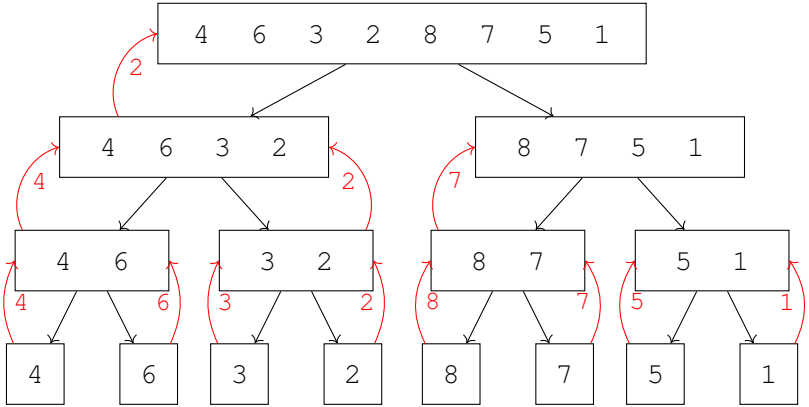

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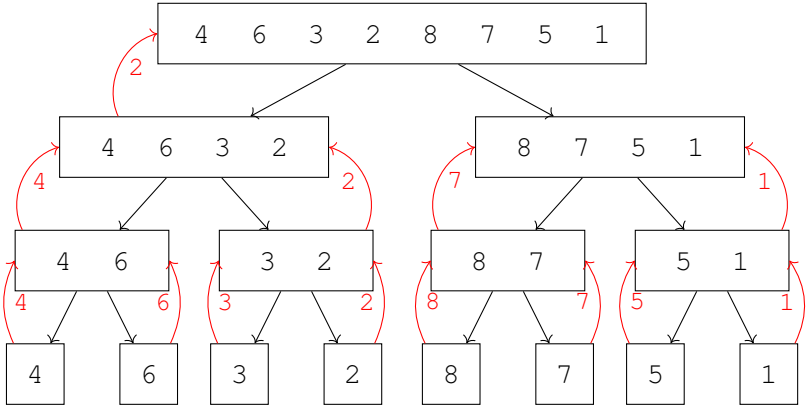

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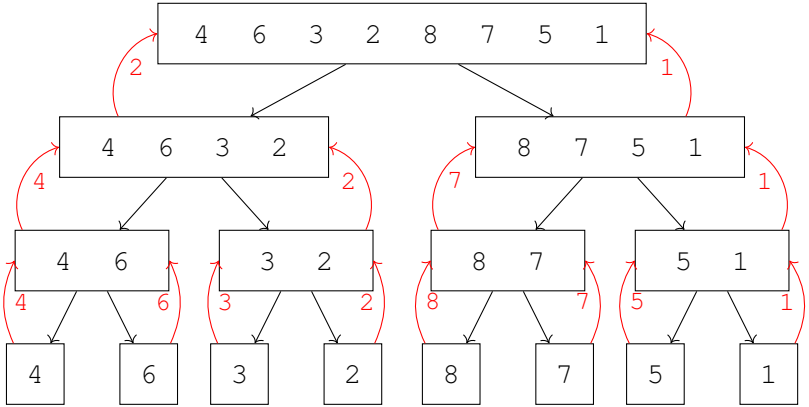

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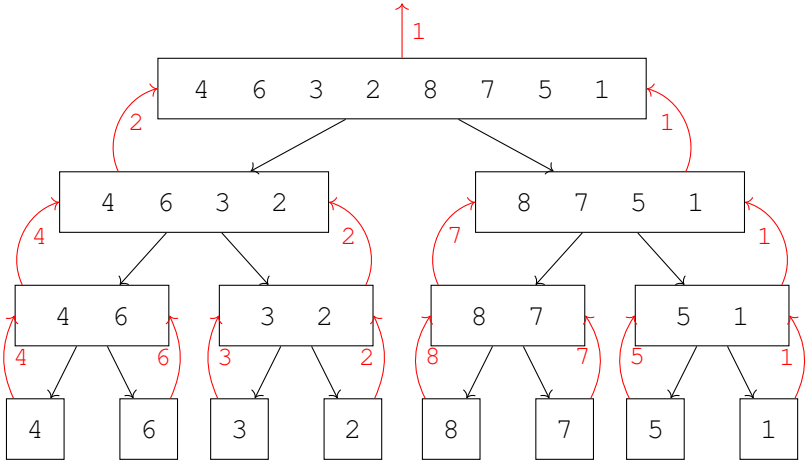
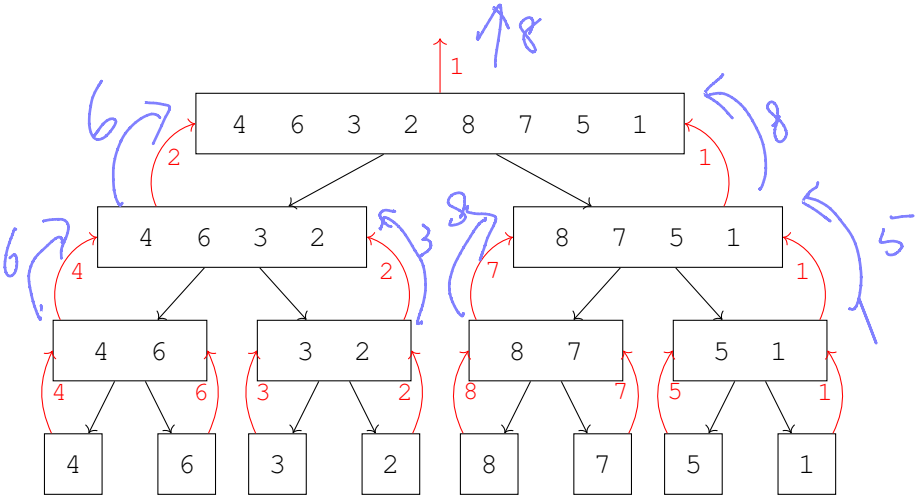


Divide-and-conquer to find the minimum

What about the maximum?



Divide-and-conquer to find the minimum

SampleRecMin.java

For simplicity, assume n is a power of 2

We can call the following algorithm by RecMin(A, 1, n)

Algorithm RecMin($A[\,]$, p, q)

Iterative version:

```
min ← A[1]
i ← 2
while i ≤ n do
begin
   if A[i] ≤ min then
      min ← A[i]
   i ← i + 1
end
output min
```

Divide-and-conquer to find the minimum

SampleRecMin.java

For simplicity, assume n is a power of 2

We can call the following algorithm by RecMin(A, 1, n)

Algorithm RecMin($A[\,]$, p, q)
    if $p == q$ then
        return $A[p]$
    else begin

    end

Iterative version:
min $\leftarrow A[1]$
i $\leftarrow$ 2
while i $\leq$ n do
begin
  if $A[i] \leq$ min then
    min $\leftarrow A[i]$
  i $\leftarrow$ i + 1
end
output min

Divide-and-conquer to find the minimum

SampleRecMin.java

For simplicity, assume n is a power of 2

We can call the following algorithm by RecMin(A, 1, n)

Algorithm RecMin($A[\,]$, p, q)
  if $p == q$ then
    return $A[p]$
  else begin
    answer1 $\leftarrow$ RecMin(A, p, $\frac{p+q-1}{2}$)
    answer2 $\leftarrow$ RecMin(A, $\frac{p+q+1}{2}$, q)

  end

Iterative version:
min $\leftarrow A[1]$
i $\leftarrow$ 2
while i $\leq$ n do
begin
  if $A[i] \leq$ min then
    min $\leftarrow A[i]$
  i $\leftarrow$ i + 1
end
output min

Divide-and-conquer to find the minimum

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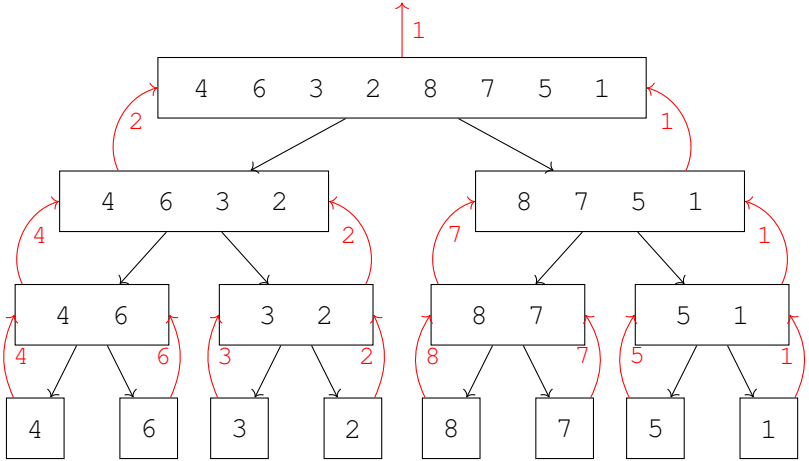
    answer2 $\leftarrow$ RecMin(A, $\frac{p+q+1}{2}$, q)
    if **answer1** $\leq$ **answer2** then
      return answer1
    else return answer2
  end

Iterative version:
min $\leftarrow A[1]$
i $\leftarrow$ 2
while i $\leq$ n do
begin
  if $A[i] \leq$ min then
    min $\leftarrow A[i]$
  i $\leftarrow$ i + 1
end
output min

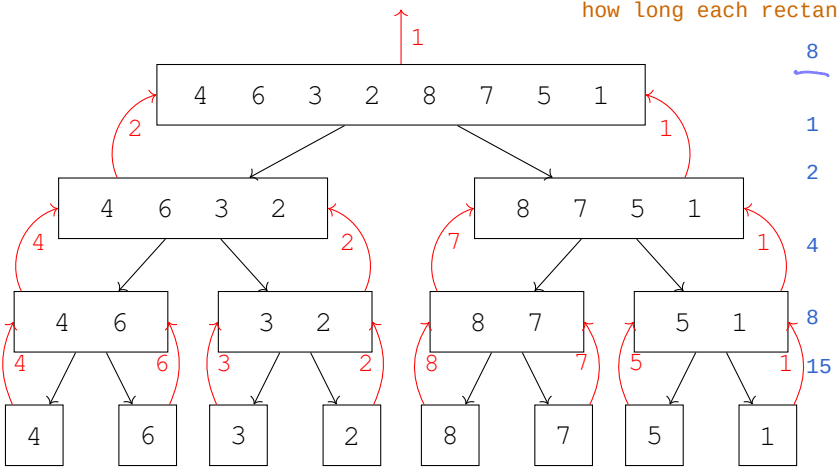
maximum??

Time complexity analysis



Time complexity analysis

how many rectangles? 2n-1
how long each rectangle take? O(1)



8 16 32
1 1 1
1 2 2
2 4 4
4 8 8
8 16 16
15 31 32
63

$O(n)$ Why?

$O(n)$

Summary

Summary: Basic Divide-and-Conquer algorithms

Next: Merge Sort Algorithm

For note taking